

Kexin Lin

Ph.D. Candidate in Business Administration | WPI Business School

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Education

Worcester Polytechnic Institute, Worcester, MA

Expected 2027

Ph.D. in Business Administration

University of Science and Technology of China, Hefei, China

2021–2023

M.Sc. in Management Science and Engineering

North China Electric Power University, China

2017–2021

B.Eng. in Engineering Cost

Research Interests

- Data envelopment analysis (DEA), performance evaluation, and benchmarking
- Machine learning and mathematical optimization for performance-driven decision support
- Performance-oriented clustering and recommendation systems for digital platforms
- Empirical modeling for sustainable operations, supply chains, and urban energy systems

Teaching Interests

- Operations management
- Business analytics
- Operations research
- Performance analytics and benchmarking
- Process modeling and optimization
- Sustainability and energy systems analytics

Publications

- **Lin, K.**, and Zhu, J. (2026). “Enhancing Deep Learning with Data Envelopment Analysis: Market-Oriented Recommendation System for Non-Fungible Tokens.” *Annals of Operations Research*. [DOI](#)
Integrates DEA-based performance scores and benchmarking relationships with deep learning to support NFT recommendations; evaluated using CryptoPunks data and comparisons with alternative recommendation methods.
- Wu, J., **Lin, K.**, and Sun, J. (2023). “Improving Urban Energy Efficiency: What Role Does the Digital Economy Play?” *Journal of Cleaner Production*, 418, 138104. [DOI](#)
- Wu, J., **Lin, K.**, and Sun, J. (2023). “Pressure or Motivation? The Effects of Low-Carbon City Pilot Policy on China’s Smart Manufacturing.” *Computers & Industrial Engineering*, 183, 109512. [DOI](#)
- Wu, J., **Lin, K.**, and Sun, J. (2025). “The Impact of Smart City Construction on Urban Energy Efficiency: Evidence from China.” *Environment, Development and Sustainability*, 27, 30337–30360. First published online in 2024. [DOI](#)

Working Papers

- **Lin, K.**, and Zhu, J. “NBDE: Nash Bargaining DEA-Enhanced Clustering Model.” *Working paper*.
Develops performance-oriented clustering using Nash-bargaining DEA weight profiles as clustering criteria. Uses Evolutionary Monte Carlo to explore cluster partitions and weighted KNN for out-of-sample assignment. Evaluation combines synthetic experiments, comparisons with alternative methods, and an application to Twitch streamers.

Conference Presentations

- “NBDE: Nash Bargaining DEA-Enhanced Clustering Model.” WPI Business School PhD Seminar, 2026.
- “Enhancing Deep Learning with Data Envelopment Analysis: Market-Oriented Recommendation System for Non-Fungible Tokens.” Decision Sciences Institute Annual Meeting, 2025; INFORMS Annual Meeting, 2025; WPI Business School PhD Seminar, 2025.

Teaching Experience

Teaching Assistant and Instructor for Selected Lessons

2024–present

Worcester Polytechnic Institute, Business School

Prepare instructional materials and deliver selected lessons on DEA and machine learning. Support undergraduate and graduate students through office hours, project guidance, grading, and feedback on analytical reasoning and interpretation of results.

- **OIE 4410:** *Case Studies in Industrial Engineering* (Undergraduate). Guest instruction on DEA; case-study project guidance, office hours, and assessment.
- **OIE 548:** *Performance Analytics* (Graduate). Support performance-analytics coursework through grading, office hours, and project guidance.
- **OIE 552:** *Modeling and Optimizing Processes* (Graduate). Guide students in problem formulation, Excel Solver implementation, scenario analysis, and interpretation of optimization results.

Recent instructional design: Developing and adapting an Excel-based regression and K-nearest neighbors (KNN) module that combines conceptual explanations, guided demonstrations, hands-on exercises, and a small independent project. Emphasizes prediction, interpretation, and analytical support for business decisions.

Honors and Awards

- Conference Travel Grant, Worcester Polytechnic Institute, 2025

Professional Service

Ad Hoc Journal Reviewer

- *Annals of Operations Research*
- *Journal of the Operational Research Society*
- *Ecological Indicators*

Technical Skills

Programming and statistical software: Python, MATLAB, R, Stata

Analytical tools: Excel and Solver

Languages

Chinese (Native), English (Fluent)